Practice workbook · Find named items in a program

Reading an opened or printed program and naming an identifier by its role: numeric/string/real variable, local or global variable, constant, data structure, user-written or built-in subprogram, parameter, argument passed, variable receiving a return value.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-12

1. Question 1 · part d.iv · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part d.iv only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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1. Question 1 · part d.iv · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part d.iv only.

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DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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1. Question 1 · part d.iv · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part d.iv only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

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Working and answer

2. Question 2 · part c.iii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iii only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

2. Question 2 · part c.iii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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2. Question 2 · part c.iii · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iii only.

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(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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2. Question 2 · part c.iii · 1 mark · Working and answer

Working and answer

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3. Question 2 · part c.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iv only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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3. Question 2 · part c.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iv only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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3. Question 2 · part c.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part c.iv only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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3. Question 2 · part c.iv · 1 mark · Working and answer

Working and answer

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4. Question 1 · part e.iii · 1 mark

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Further practice · Attempt part e.iii only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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4. Question 1 · part e.iii · 1 mark

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Further practice · Attempt part e.iii only.

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(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

4. Question 1 · part e.iii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part e.iv · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part e.iv only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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5. Question 1 · part e.iv · 1 mark

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Further practice · Attempt part e.iv only.

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(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

5. Question 1 · part e.iv · 1 mark · Working and answer

Working and answer

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6. Question 3 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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6. Question 3 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

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(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

Figure 1 shows a Vigenere square.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

(4)

Encrypted letter	f	l	m	k	t	r	w	h	e	e
Keyword letter										
Decrypted letter										

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6. Question 3 · part d.ii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.ii only.

(d) A user-defined function uses a Vigenere cipher to encrypt plaintext into ciphertext.

This pseudocode shows how the function is called.

```
SET ciphertext TO ciphertext & getCipherLetter(keywordLetter, plaintextLetter)
```

(i) State what would be returned by this function.

(1)

.....

.....

(ii) Give the name of **one** parameter passed to this function.

(1)

.....

(iii) State what is meant by a **built-in subprogram**.

(1)

.....

.....

(Total for Question 3 = 17 marks)

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Working and answer

7. Question 3 · part a.i · 1 mark

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Further practice · Attempt part a.i only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

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7. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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8. Question 3 · part a.ii · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.ii only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

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9. Question 3 · part a.iii · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.iii only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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9. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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10. Question 3 · part a.iv · 1 mark

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Further practice · Attempt part a.iv only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA

10. Question 3 · part a.iv · 1 mark · Working and answer

Working and answer

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11. Question 3 · part a.v · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a.v only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

11. Question 3 · part a.v · 1 mark · Working and answer

Working and answer

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12. Question 2 · part a.iv · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iv only.

2 Raza is writing a program to tell the user whether a number they input is a prime number.

A prime number is a whole number, larger than one, that can only be divided by one and itself with no remainder.

This pseudocode contains the logic required to complete the program.

```
FUNCTION checkPrime(pNumber)
BEGIN FUNCTION
    IF pNumber = 1 THEN
        check = False
    ELSE
        check = True
        FOR count FROM 2 TO pNumber DO
            IF pNumber MOD count = 0 THEN
                check = False
            END IF
        END FOR
    END IF
RETURN check
END FUNCTION

SEND "Enter a number: " TO DISPLAY
RECEIVE number FROM (INTEGER) KEYBOARD
SET result TO checkPrime(number)

IF result = True THEN
    SEND (number & " is a prime number") TO DISPLAY
ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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12. Question 2 · part a.iv · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.iv only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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12. Question 2 · part a.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 2 · part a.v · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.v only.

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This pseudocode contains the logic required to complete the program.

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IF result = True THEN
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ELSE
    SEND (number & " is not a prime number") TO DISPLAY
END IF
```

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

13. Question 2 · part a.v · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a.v only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Answer these questions about the pseudocode.

(i) Identify a line number where **selection** starts.

(1)

(ii) Identify the line number where **iteration** starts.

(1)

(iii) Identify a line number where a **string** and a **number** are printed on the same line.

(1)

(iv) Identify the name of a **local variable**.

(1)

(v) Identify the name of a **parameter**.

(1)

(b) Write a program to implement the logic in the pseudocode.

Open **Q02b** in the code editor.

Write the program.

You must use the structure given in **Q02b** to complete the program.

Do not add any further functionality.

Save your code as **Q02bFINISHED** with the correct file extension for the programming language.

(11)

(Total for Question 2 = 16 marks)

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13. Question 2 · part a.v · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iii only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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14. Question 1 · part b.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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14. Question 1 · part b.iii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 1 · part b.iv · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iv only.

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(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
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04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
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29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

15. Question 1 · part b.iv · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA

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Use the code to answer these questions.

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(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

OLevels.com SCHOOL

15. Question 1 · part b.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 1 · part b.v · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.v only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16. Question 1 · part b.v · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part b.v only.

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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16. Question 1 · part b.v · 1 mark · Working and answer

Working and answer

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17. Question 1 · part d.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.i only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

17. Question 1 · part d.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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17. Question 1 · part d.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.i only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN

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Working and answer

18. Question 1 · part d.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.iii only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

18. Question 1 · part d.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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18. Question 1 · part d.iii · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part d.iii only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN

OLevels.COM SCHOOL

Working and answer